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Chapter 5 — Functions of One Random Variable

5-3 Mean and Variance

Expected Value (Mean)

- **Definition:** The expected value or mean of a continuous random variable x is defined as the integral $\eta_x = E\{x\} = \int_{-\infty}^{\infty} xf(x)dx$.
- **Physical Interpretation:** If the probability density function (PDF) $f(x)$ is interpreted as a mass density on the x -axis, the expected value $E\{x\}$ corresponds to its center of gravity.
- **Discrete Type:** For a discrete random variable taking values x_i with probabilities P_i , the mean is the sum $E\{x\} = \sum_i P_i x_i$.
- **Conditional Mean:** The conditional mean assuming an event M replaces $f(x)$ with the conditional density $f(x|M)$, yielding $E\{x|M\} = \int_{-\infty}^{\infty} xf(x|M)dx$.

Mean of a Function $g(x)$

- **Fundamental Theorem:** To find the mean of a new random variable $y = g(x)$, one does not need the density of y . Instead, it can be computed directly using the density of x :

$$E\{g(x)\} = \int_{-\infty}^{\infty} g(x)f(x)dx$$

- **Linearity:** The expected value is a linear operator. For constants a_i and functions $g_i(x)$, it holds true even if $g(x)$ is complex:

$$E\{a_1g_1(x) + \cdots + a_ng_n(x)\} = a_1E\{g_1(x)\} + \cdots + a_nE\{g_n(x)\}$$

Variance

- **Definition:** The variance σ^2 measures the concentration or spread of a random variable around its mean η . It is defined as the expected value of the squared deviation:

$$\sigma^2 = \text{var}(x) = E\{(x - \eta)^2\} = E\{x^2\} - (E\{x\})^2$$

- **Standard Deviation:** The positive square root of the variance, σ , is the standard deviation, representing the root-mean-square (RMS) value of the random variable x around its mean η .
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5-4 Moments

Types of Moments

- **General Moments:** The n -th moment is $m_n = E\{x^n\} = \int_{-\infty}^{\infty} x^n f(x) dx$.
- **Central Moments:** The n -th central moment measures deviation from the mean: $\mu_n = E\{(x - \eta)^n\}$. Note that $\mu_0 = m_0 = 1$, $\mu_1 = 0$, and $\mu_2 = \sigma^2$.
- **Absolute Moments:** Defined as $E\{|x|^n\}$.

Estimates of Mean and Variance of $g(x)$

When x is highly concentrated near its mean η , we can estimate the moments of $y = g(x)$ using a Taylor series expansion of $g(x)$ around η : * **Mean Estimate:** $\eta_y = E\{g(x)\} \approx g(\eta) + g''(\eta) \frac{\sigma^2}{2}$. * **Variance Estimate:** $\sigma_y^2 \approx |g'(\eta)|^2 \sigma^2$.

Fundamental Inequalities

- **Markov Inequality:** If $x \geq 0$ (a non-negative random variable) and $a > 0$:

$$P\{x \geq a\} \leq \frac{E\{x\}}{a}$$

- **Chebyshev (Tchebycheff) Inequality:** Bounds the probability that x deviates from its mean η by more than ϵ :

$$P\{|x - \eta| \geq \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}$$

- This fundamental result holds for *any* density $f(x)$ and is used when the exact density is unknown.
 - **Bienaymé Inequality:** For an arbitrary number a and integer n : $P\{|x - a| \geq \epsilon\} \leq \frac{E\{|x - a|^n\}}{\epsilon^n}$.
 - **Lyapunov Inequality:** Relates absolute moments $P_k = E\{|x|^k\}$, demonstrating that $(P_{k-1})^{1/(k-1)} \leq (P_k)^{1/k}$.
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5-5 Characteristic Functions

Definition and Properties

- **Characteristic Function:** Defined as the expectation of the complex exponential $e^{j\omega x}$, which is equivalent to the Fourier transform of the probability density function $f(x)$ (with a positive sign in the exponent):

$$\Phi_x(\omega) = E\{e^{j\omega x}\} = \int_{-\infty}^{\infty} f(x) e^{j\omega x} dx$$

- **Maximum at Origin:** The magnitude is bounded by its value at the origin: $|\Phi_x(\omega)| \leq \Phi_x(0) = 1$.
- **Inversion Formula:** The density $f(x)$ can be uniquely recovered via the inverse Fourier transform:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi_x(\omega) e^{-j\omega x} d\omega$$

Moment Generating Function

- **Definition:** By replacing $j\omega$ with a real variable s , we obtain the moment generating function: $\Phi(s) = E\{e^{sx}\} = \int_{-\infty}^{\infty} f(x)e^{sx}dx$.
- **Moment Theorem:** The n -th derivative of the moment function evaluated at $s = 0$ yields the n -th general moment of x :

$$\Phi^{(n)}(0) = E\{x^n\} = m_n$$

- **Series Expansion:** If all moments are finite, the moment function can be expanded as $\Phi(s) = \sum_{n=0}^{\infty} \frac{m_n}{n!} s^n$. Thus, a random variable's density is uniquely determined if all its moments are known.

Cumulants

- **Definition:** The cumulants λ_n of a random variable are the derivatives of the *second* moment function (the natural logarithm of the moment generating function, $\Psi(s) = \ln \Phi(s)$) evaluated at the origin:

$$\lambda_n = \left. \frac{d^n \Psi(s)}{ds^n} \right|_{s=0}$$

Discrete-Type Random Variables

- **Z-Transform Method:** For a discrete random variable taking integer values n with probabilities P_n , the moment generating function is defined using the z-transform:

$$\Gamma(z) = E\{z^x\} = \sum_{n=-\infty}^{\infty} P_n z^n$$

- **Factorial Moments:** Differentiating the z-transform k times and evaluating at $z = 1$ yields factorial moments:

$$\Gamma^{(k)}(1) = E\{x(x-1)\cdots(x-k+1)\}$$